Fiscal Year:	FY 2009	**Task Last Updated:**	FY 04/08/2010
PI Name:	Bhattacharya, Sharmila		
Project Title:	Fungal Pathogenesis, Tumorigenesis, and Effects of Host Immunity in the Space Environment (FIT Flight Experiment)		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Musculoskeletal Biology (4) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	Sharmila.Bhattacharya@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	
Organization Name:	NASA Ames Research Center		
PI Address 1:	Building: N236, Room: 213		
PI Address 2:	MS 236-5		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	14
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Directed Research
Start Date:	09/01/2005	**End Date:**	08/31/2009
No. of Post Docs:	4	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-121 NOTE: End date should be 8/31/2009, per PI and A. Chu/ARC (4/2010) NOTE: Received NCE to 8/31/2008 per A. Chu/ARC (9/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kimbrell, Deborah (U CA Davis)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Fungal Pathogenesis, Tumorigenesis, and Effects of Host Immunity in the Space Environment (FIT Flight Experiment): The FIT flight experiment used a well-defined biological model system, D. melanogaster to study how space flight affects immune system function. The goal of the FIT experiment was to collect data on the cellular and molecular biological effects of the space flight environment on immune system response to bacterial and fungal infections.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A better understanding the effects of pathogens on wild type and immune-compromised hosts is of great relevance to human immune function and diseases.
Task Progress:	The FIT experiment (Fungal Pathogenesis, Tumorigenesis, and Effects of Host Immunity in the Space Environment) on STS-121 was designed to assess the effects of spaceflight on the immune system response of a host, the fruit fly Drosophila melanogaster, as a model for the innate immune responses of higher organisms. Both the microgravity environment and the re-adaptation to 1g upon return of the experiment to Earth affected the overall immunity of the flies (expression profiles of immune genes), as well as the specific cellular (number and activity of phagocytosing cells) and humoral response (antimicrobial peptide production) upon infection with E.coli pathogen. The overall ability to fight bacterial infections reflected the combined genetic and physiological changes due to exposure to the microgravity environment. The experiment also confirmed previous findings of developmental alterations in space, and uncovered changes in the expression level of genes involved in cell division, reproduction and oogenesis. Similar to results from cells grown in culture in microgravity or simulated microgravity, the fly metabolism was also altered by spaceflight. Both the developmental and metabolic changes may contribute to the smaller population size that survived spaceflight, while the immune system alteration is an important factor in determining survival to bacterial infections upon return to Earth.
Bibliography Type:	Description: (Last Updated: 03/02/2021)
Abstracts for Journals and Proceedings	Bhattacharya S, Marcu O. "Space Flight Experiments." Invited talk at the 55th Annual Meeting of the Radiation Research Society, Savannah, Georgia, October 3-7, 2009. Proceedings of the 55th Annual Meeting of the Radiation Research Society. #S1601:p. 28, October 2009. , Oct-2009
Abstracts for Journals and Proceedings	Elland CS, Navarro BJ, Fernandez J, Day BH, Sato K, Bhattacharya S, Bulkowski J. "Flies in Space website." Presented at the 22nd ASBSB annual meeting, Arlington, VA, November 2-5, 2006. Gravitational and Space Biology. 2006 Nov;20(1):30. , Nov-2006
Articles in Peer-reviewed Journals	Inan OT, Etemadi M, Sanchez ME, Marcu O, Bhattacharya S, Kovacs GT. "A miniaturized video system for monitoring the locomotor activity of walking Drosophila melanogaster in space and terrestrial settings." IEEE Trans Biomed Eng. 2009 Feb;56(2):522-4. Epub 2008 Oct 3. PMID: 19272912 , Feb-2009
Articles in Peer-reviewed Journals	Fahlen T, Sanchez M, Lera M, Blazevic E, Chang J, Bhattacharya S. "A study of the effects of space flight on the immune response in Drosophila melanogaster." Gravitational and Space Biology 2006 Aug;19(2):133-4. http://gravitationalandspacebiology.org/index.php/journal/article/viewFile/20/20 , Aug-2006
Articles in Peer-reviewed Journals	Marcu O, Lera MP, Sanchez ME, Levic E, Higgins LA, Shmygelska A, Fahlen TF, Nichol H, Bhattacharya S. "Innate immune responses of Drosophila melanogaster are altered by spaceflight." PLoS One. 2011 Jan 11;6(1):e15361. http://dx.doi.org/10.1371/journal.pone.0015361 , Jan-2011
Dissertations and Theses	Selch F. "Apoptosis and Immune System Adaptation in Drosophila melanogaster following Space Flight and Irradiation." Dissertation. Jointly with NASA Ames Research Center and the University of Vienna, Austria. September 2007. , Sep-2007